

1 Week 1

1.1 Paradox

A declarative statement that cannot be true and cannot be false.

1.2 Compact definition

A natural number n has a compact definition if there is an English sentence of at most 200 characters that uniquely define n .

1.3 Proposition

A declarative sentence that is either true or false in some context

1.4 Atomic proposition (atom)

A proposition that cannot be broken down into smaller propositions. A proposition that is not atomic is called **compound**

1.5 Unique Readability of Propositional Formulas

For every propositional formula in $\text{Form}(\mathcal{L}^p)$, A is exactly an atom, $(\neg B)$, $(B \wedge C)$, $(B \vee C)$, $(B \rightarrow C)$ or $(B \leftrightarrow C)$ and in each case A is formed in exactly one way.

1.6 Proper Initial Segment

A proper initial segment of A is a non empty expression X such that A is XY for some non empty expression Y

1.7 Unique Readability Theorem

- The first symbol of A is either (or a variable.
- A has an equal number of (and) and each proper initial segment of A has more (than)
- A has a unique construction as a formula.

1.8 Order of Precedence

1. $\neg$
2. $\wedge$
3. $\vee$
4. $\rightarrow$
5. $\leftrightarrow$

2 Week 2

2.1 Truth Valuation

A truth valuation, t , is a function from propositional symbols to $\{0, 1\}$. p^t denotes the value of p under t .

$$t : P \rightarrow \{0, 1\}$$

where P is the set of available propositional symbols

2.2 Satisfied

C is satisfied under t if $C^t = 1$ and not satisfied if $C^t = 0$

2.3 Tautologically Equivalent

If $A^t = B^t$ for every truth valuation, t . ($A \models B$)

2.4 Tautology

It is a tautology or a valid formula if $C^t = 1$ for every truth valuation t .

2.5 Satisfiable

A propositional formula C is satisfiable if $C^t = 1$ for some truth valuation t .

2.6 Contradiction / Not Satisfiable

If every truth valuation $C^t = 0$

2.7 Set Satisfiable

t satisfies a set Σ if for every $C \in \Sigma$, $C^t = 1$. If there exist $C \in \Sigma$ such that $C^t = 0$, t does not satisfy Σ . ($\Sigma^t = 0$)

2.8 Tautologically Implies

Σ tautologically implies a propositional formula C ($\Sigma \models C$) if whether $\Sigma^t = 1$ for some truth valuation t , we also have $C^t = 1$. We say that C is a **tautological consequence** of Σ

3 Week 3

3.1 Replacability Theorem

Assume $B \models C$, and let A' be the formula obtained by replacing some of A with occurrences of B with formula C . Then $A' \models A$.

3.2 Duality Theorem

Suppose A is a formula formed with only atoms, $\neg, \wedge, \vee$. Suppose $\Delta(A)$ replaces A with all occurrences of $\wedge$ with $\vee$ and $\vee$ with $\wedge$, as well as each atom with its negation. Then $\neg A \models \Delta(A)$

3.3 Literal

Either a propositional symbol, or negation of a propositional symbol.

3.4 Disjunctive Clause

$$(p \vee (\neg q) \vee r)$$

3.5 Conjunctive Clause

$$((\neg p) \wedge q \wedge (\neg r))$$

3.6 Disjunctive Normal Form (DNF)

$$(A \wedge \dots \wedge B) \vee \dots \vee (C \wedge \dots \wedge D)$$

3.7 Conjunctive Normal Form (CNF)

$$(A \vee \dots \vee B) \wedge \dots \wedge (C \vee \dots \vee D)$$

4 Week 4

4.1 Arity

The number of inputs the function takes.

1-ary function ($\neg$) are called **unary**

2-ary function ($\wedge, \vee \dots$) are called **binary**

Arity n function: $f : \{0, 1\}^n \rightarrow \{0, 1\}$

4.2 Adequate for Propositional Logic

If every propositional connectives can be implemented using the connectives in S .

4.3 Theorems for Propositional Logic

The set $\{\wedge, \vee, \neg\}$ is an adequate set of connectives for propositional logic. Same for the sets $\{\wedge, \neg\}, \{\vee, \neg\}, \{\rightarrow, \neg\}$.

4.4 Dead Code

Code that is never executed.

5 Week 5

5.1 Formula Deduction Proof

It is a sequence of lines of the form $\Sigma \vdash A$ (Σ proves A) for some set Σ and formula A .

5.2 Soundness of Propositional Formula Deduction

$$\Sigma \vdash C \implies \Sigma \models C$$

5.3 Consistent

We call Σ **consistent**

- If for every propositional formula A , if $\Sigma \vdash A$ then $\Sigma \not\vdash (\neg A)$ or,
- If there exists formula B such that $\Sigma \not\vdash B$

5.4 Inconsistent

We call Σ **inconsistent** if it is not consistent, so if Σ is inconsistent, for every propositional formula C ,

$$\Sigma \vdash C, \Sigma \vdash (\neg C)$$

5.5 Lemma

$$\Sigma \models A \iff \Sigma \cup \{(\neg A)\} \text{ is unsatisfiable.}$$

$$\Sigma \vdash A \iff \Sigma \cup \{(\neg A)\} \text{ is inconsistent.}$$

5.6 Completeness of Propositional Formula Deduction

$$\Sigma \models A \implies \Sigma \vdash A$$

If Σ is consistent, then it is satisfiable.

If Σ is satisfiable, then it is consistent.

6 Week 6

6.1 Refutation System

In a **refutation system** to prove the argument $A_1, \dots, A_n \models C$ is valid, we show that the set

$$\{A_1, \dots, A_n, \neg C\}$$

is not satisfiable or inconsistent.

We show that $\{A_1, \dots, A_n, \neg C\}$ can formally prove both B and $\neg B$ for some formula B .

6.2 Formal derivation

Formal derivation (formal deduction) has the following features:

- Only formulas allowed are the disjunction of literals e.g. $(p \vee q \vee \neg r)$ (Disjunctive clauses)
- Follows the refutation principle, showing that the negation of the conclusion is inconsistent

In general, we convert any formula into one or more disjunctive clauses. To do this, we convert the formula into a conjunction of disjunction, converting the formulas into CNF. Each clause of the conjunction is an input formula for resolution. e.g. we convert $p \rightarrow (q \wedge r) \models (\neg p \vee q) \wedge (\neg p \vee r)$

6.3 Resolution

$$C \vee p, D \vee \neg p \vdash_r C \vee D$$

$C \vee p, D \vee \neg p$ are **parent clauses**, we say we are resolving two parent clauses over p .

$C \vee D$ is the **resolvent**.

The resolvent of p and $\neg p$ is called the empty clause ($\perp$)

$$p, \neg p \vdash_r \perp$$

A **(resolution) derivation** from a set of clauses $\mathcal{S}$ is a finite sequence of clauses such that each clause is either in $\mathcal{S}$ or results from previous clauses in the sequence by resolution.

Remarks:

- Clauses can only be resolved iff they contain two complementary literals $(p, \neg p)$
- The empty clause $\perp$ is not satisfiable.

e.g.

Need formulas A, B written as some disjunction of literals

$$A = p \vee q \vee \neg r$$

$$B = \neg p \vee r$$

To apply the resolution proof rule to A, B , there must exist a propositional symbol which occurs in A with its negation in B ($p, \neg p, r, \neg r$).

Let's work with p . Let A as $p \vee C$, B as $\neg p \vee D$ for subformulas C, D dictated by A, B . The resolvent of A, B is $C \vee D$ after we resolve p with $\neg p$. Therefore, $C = q \vee \neg r$, $D = r$, $C \vee D = q \vee \neg r \vee r$

6.4 Possible outcomes from Resolution Refutation

1. Arrive at empty clause ($\perp$). (not satisfiable, by soundness, implies the original clause is not satisfiable. Hence, the original argument is valid)
2. Resolve everything possible without obtaining $\perp$. This suggests starting clause might be satisfiable. Hence, the original argument was not valid. Confirm by finding a satisfiable truth valuation for the set.

6.5 Soundness of Resolution

This resolvent is tautologically implied by its parent clause, resolving a sound rule of formal deduction.

Corollary: Resolution preserves the satisfiability of the set at every step.

6.6 Set of Support strategy

1. Partition all clauses into 2 sets, the set of support (negation of conclusion) and auxiliary set (premise)
2. Auxiliary set is formed in a way that formulas in it are **not contradictory**. The contradiction will only arise when one adds the negation of the conclusion.
3. Resolve clauses within the auxiliary set to avoid contradiction.
4. **Rule:** every resolution step must use at least one formula from the set of support.
5. Resolvent is added to the set of support
6. **Theorem:** resolution with the set of support strategy is complete.

6.7 David-Putnam Procedure

- Any clause corresponds to a set of literals. e.g. $p \vee \neg q \vee r$ corresponds to $\{p, \neg q, r\}$
- Orders of disjunction is irrelevant, and duplicates don't matter, the set associated with the clause determines the clause
- We treat clauses as sets, allows one to speak of union of two clauses
- If clauses are represented as sets, we write the resolvents on p

$$(C \cup \{p\}) \cup (D \cup \{\neg p\} \setminus \{p, \neg p\})$$

- By defn, $\{p\}$ and $\{\neg\}$ is empty clause

$$C = (A \cup B) \setminus \{p, \neg p\}$$